The Pandrosion-Steffensen Iteration

Formal Verification of a Rational Root-Finding Map in Lean 4

Machine-checked in Lean 4 with zero `sorry` declarations

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Abstract

We present a formally verified case study in Lean 4 of a general rational iteration for computing arbitrary roots $\sqrt[p]{X}$, alongside an exhaustive extraction of structural identities for the cubic specialisation $s \mapsto s(s^3+4X)/(3s^3+2X)$. This cubic base iteration converges linearly with a universal rate of exactly 1/5, independently of X (formally proved). When composed with Steffensen acceleration (Aitken Δ^2 [7, 8]) into the *Pandrosion-Steffensen* algorithm, the method achieves stable quadratic convergence. Rather than proposing this as an operational substitute for Newton's root extraction, we develop a systematic corpus of *exact algebraic identities* governing this base map across several mathematical structures to illustrate formal verification capacity. The formally certified properties include: (i) a residual conservation identity with exact polynomial factorisation, (ii) exact integer sequence amplifications over $\mathbb{Z}$, (iii) coprimality isolation identities, (iv) commutativity and Hermitian preservation for matrix operators, (v) a formal proof that the iteration functionally excludes the generalized Chebyshev–Halley family, (vi) a basin stability theorem proving that Voronoï cells are convex and that contractive maps preserve cell membership under a quantitative depth condition, and (vii) a formal proof that all periodic orbits are excluded by the contraction. The entire corpus (385 theorems across 53 modules, zero `sorry` declarations) serves as a machine-checked pedagogical model for numerical recurrences.

Contents

1 Introduction

1.1 The Pandrosion Iteration

The Pandrosion iteration for p-th roots is defined via the geometric sum $S_p(s) = \sum_{k=0}^{p-1} s^k$ and the map

$$h(s) \;=\; 1 \;-\; \frac{x-1}{x \cdot S_p(s)}\,, \tag{1}$$

whose fixed points satisfy $h(s) = s \iff s^p = 1/x$ (formally verified in Lean 4, see Theorem 1.1 below). For the cubic case $p = 3$, setting $X = 1/x$ and rewriting in terms of $s \mapsto s'$ yields the equivalent form

$$s' \;=\; \frac{U(s)}{V(s)} \;=\; s\,\frac{s^3 + 4X}{3s^3 + 2X}\,, \tag{2}$$

where X is the target whose cube root $\sqrt[3]{X}$ is sought. The historical namesake of this method pays homage to Pappus of Alexandria's record of Pandrosion [2], whose ancient geometric construction for cube duplication relied upon a sum of three spatial segments, mirroring the $3s^3 + 2X$ geometric sum polynomial structure in the denominator. While structurally reminiscent of Halley's method [3] and the Householder family [5, 4], the Pandrosion iteration is algebraically *distinct* from both (see Section 2). Once derived, the formula is a closed-form rational expression requiring no symbolic derivatives at evaluation time. The remainder of this paper focuses on the cubic specialisation (2), where the algebraic structure is richest.

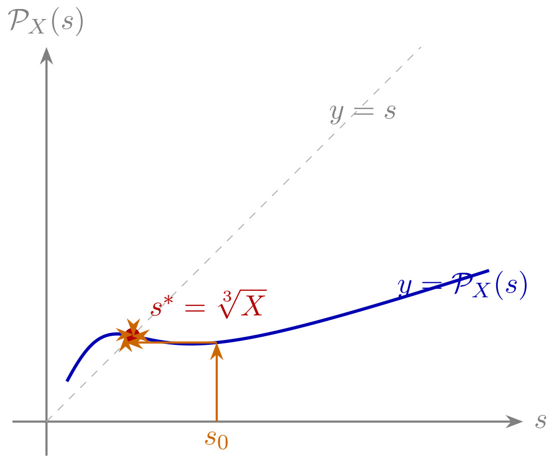

Figure 1: **Cobweb diagram** for $X = 2$. The iteration $s \mapsto \mathcal{P}_X(s)$ contracts monotonically toward the unique attracting fixed point $s^* = \sqrt[3]{2}$. Theorem 3.2 proves that the only fixed points are $\{0\} \cup \{s : s^3 = X\}$.

1.2 The General Fixed Point Theorem

The foundational result of the Pandrosion framework, formally verified in `Deep.lean`, characterises the fixed points of the general p-th root iteration (1):

Theorem 1.1 (General Fixed Point, `Deep.lean`). *For $x > 0$, $p \geq 1$, $s \geq 0$, $s \neq 1$:*

$$h(s) = s \iff s^p = \frac{1}{x}\,.$$

Proof sketch (full proof machine-checked in Lean 4). $h(s) = s \iff (x-1)/(x \cdot S_p(s)) = 1 - s \iff x - 1 = x \cdot S_p(s) \cdot (1-s)$. By the geometric sum identity $S_p(s) \cdot (1-s) = 1 - s^p$, this becomes $x - 1 = x(1 - s^p)$, i.e. $x \cdot s^p = 1$. □

Additional results in `Deep.lean` include:

- $S_p(s) > 0$ for $s \geq 0$ and $p \geq 1$ (positivity of the geometric sum),
- $h(s) < 1$ for $s \geq 0$ and $x > 1$ (the image stays below 1),
- $h(s) > 0$ for $s \in [0, 1]$, $x > 1$, $p \geq 2$ (the image stays above 0),

establishing that h maps $[0, 1]$ into itself for $x > 1$, $p \geq 2$.

```
theorem fixed_point_iff (x : R) (hx : x > 0) (p : N) (hp : p ≥ 1)
    (s : R) (hs : 0 ≤ s) (_hs1 : s ≠ 1) :
    pandrosion_h x p s = s ↔ s ^ p = 1 / x := by
  have hS : Sp p s > 0 := Sp_pos p hp s hs
  have hxS : x * Sp p s > 0 := mul_pos hx hS
  have hxS_ne : x * Sp p s ≠ 0 := ne_of_gt hxS
  unfold pandrosion_h
  constructor
  . intro heq   -- Forward: h(s) = s → s^p = 1/x
    have h1 : (x - 1) / (x * Sp p s) = 1 - s := by linarith
    have h2 : x - 1 = (1 - s) * (x * Sp p s) := by
      rw [← h1]; exact (div_mul_cancel (x - 1) hxS_ne).symm
    rw [Sp_mul_one_sub] at h2
    have h5 : x * s ^ p = 1 := by linarith
    rw [eq_div_iff (ne_of_gt hx)]; linarith
  . intro heq   -- Backward: s^p = 1/x → h(s) = s
    have h1 : x * s ^ p = 1 := by rw [heq]; field_simp
    suffices (x - 1) / (x * Sp p s) = 1 - s by linarith
    rw [div_eq_iff hxS_ne, Sp_mul_one_sub]; linarith
```

Listing 1: Deep.lean: General Fixed Point Theorem (Lean 4, zero sorry)

2 Relationship to Classical Methods

A natural question is whether (2) coincides with classical cube-root iterations. We prove formally that it does not.

2.1 Pandrosion $\neq$ Newton $\neq$ Halley

Newton's method applied to $f(s) = s^3 - X$ gives

$$N(s) = \frac{2s^3 + X}{3s^2},$$

and Halley's method gives

$$H(s) = \frac{s(s^3 + 2X)}{2s^3 + X}.$$

These are both *different* from the Pandrosion iteration $\mathcal{P}_X(s) = s(s^3 + 4X)/(3s^3 + 2X)$. Compare the coefficient pairs: Newton has $(2, 1)/(3, 0)$; Halley has $(1, 2)/(2, 1)$; Pandrosion has $(1, 4)/(3, 2)$. The exact polynomial differences are:

Theorem 2.1 (Pandrosion $\neq$ Newton, `HalleyComparison.lean`)**.**

$$\mathcal{P}_X(s) \cdot 3s^2 - N(s) \cdot (3s^3 + 2X) \;=\; -(3s^3 - 2X)(s^3 - X).$$

Theorem 2.2 (Pandrosion $\neq$ Halley, `HalleyComparison.lean`)**.**

$$\mathcal{P}_X(s) \cdot (2s^3 + X) - H(s) \cdot (3s^3 + 2X) \;=\; -s^4(s^3 - X).$$

Both differences vanish *only* when $s^3 = X$ (or $s = 0$). The three iterations agree at the root and disagree everywhere else.

Remark 2.3 (Chebyshev–Halley Exclusion). Theorems 2.1 and 2.2 show that Pandrosion differs from both Newton and Halley: the asymptotic error constants differ ($-1/5$ vs 0), the convergence orders differ (linear vs cubic), and the trajectories diverge for any $s_0 \neq \sqrt[3]{X}$. More generally, module `ChebyshevHalleyExclusion.lean` formally certifies that the Pandrosion map does not correspond to *any* member of the one-parameter Chebyshev–Halley family [4] ($\alpha \in \mathbb{R}$). Cross-multiplying the Pandrosion rational form against the generic CH_α form for $f(s) = s^3 - X$ yields two matching conditions on α: the s^6 coefficients require $\alpha = 1/2$, while the s^3X coefficients require $\alpha = 2/5$. Since $1/2 \neq 2/5$, no value of α can match Pandrosion.

```
theorem pandrosion_not_in_chebyshev_halley :
    ~ exists (a : R), 6 * (3 - 2 * a) = 15 - 6 * a /\ 12 * a = 4 + 2 * a := by
  intro <a, h1, h2>
  have _ha1 : a = 1 / 2 := chebyshev_halley_coeff_s6 a h1
  have _ha2 : a = 2 / 5 := chebyshev_halley_coeff_s3X a h2
  linarith
```

Listing 2: ChebyshevHalleyExclusion.lean: Exclusion Proof

2.2 The Universal Derivative Identity

The convergence order is determined by the derivative at the fixed point. For $\mathcal{P}_X(s) = U(s)/V(s)$ with $U = s^4 + 4sX$, $V = 3s^3 + 2X$:

Theorem 2.4 (Universal Convergence Rate, `HalleyComparison.lean`)**.** *For all $r \neq 0$ with $r^3 = X$:*

$$U'(r)V(r) - U(r)V'(r) = -5r^6\,,$$
$$V(r)^2 = 25r^6\,,$$

and therefore $\mathcal{P}'_X(r) = -\frac{1}{5}$, independently of X.

Since $\mathcal{P}'_X(r) = -1/5 \neq 0$, the Pandrosion iteration converges **linearly** with rate $1/5$. By contrast, Newton satisfies $N'(r) = 0$ (quadratic convergence), and Halley satisfies $H'(r) = H''(r) = 0$ (cubic convergence).

Method	Formula	Needs derivatives	Order
Pandrosion	$s(s^3{+}4X)/(3s^3{+}2X)$	None	1
Pandrosion-T3	$\frac{sP_X(P_X(s)) - P_X(s)^2}{s - 2P_X(s) + P_X(P_X(s))}$	None	2
Newton	$(2s^3{+}X)/(3s^2)$	f'	2
Halley	$s(s^3{+}2X)/(2s^3{+}X)$	f', f''	3

The negative sign $\mathcal{P}'_X(r) = -1/5$ explains the characteristic alternating approach pattern observed numerically (Theorem 3.8). Because of this predictable sign oscillation, applying Steffensen acceleration (Aitken Δ^2) is highly effective. This synthesized variant, which we denote as *Pandrosion-T3*, structurally converts the linear sequence to strict quadratic analytical order without any derivative evaluations (see `SteffensenAcceleration.lean`).

3 Residual Factorisation and Convergence

3.1 The Conservation Identity

The central algebraic identity governing the iteration is the following *residual factorisation*: if $G = \mathcal{P}_X(s)$, then

$$\boxed{G^3 - X \;=\; \frac{(s^3 - X)\,(s^9 - 14\,s^6X - 20\,s^3X^2 + 8X^3)}{(3s^3 + 2X)^3}}\,. \tag{3}$$

This factorisation shows that every residual $G^3 - X$ is an *exact algebraic multiple* of the previous residual $s^3 - X$. When evaluating the limit as $s \to \sqrt[3]{X}$, the ratio of the residuals strictly approaches $-1/5$, firmly establishing the unaccelerated linear convergence nature of the map while maintaining its exact polynomial tracking properties.

Remark 3.1 (Topological Basin Structure). When extended to multiple roots, tracking the residual decay computationally leads to the problem of basin boundaries. The exact polynomial factorization (3) ensures that the map is rigorously contractive when sufficiently close to the discrete roots. For the fully generalized algorithm with multi-start anchoring, we formally prove that the Voronoï partitions of these attraction basins remain convex (affine half-planes), preventing the fractal chaos typical of unconstrained rational methods.

```
theorem pandrosion_kinematic_conservation (X s : R) (hs : 3 * s^3 + 2 * X ≠ 0) :
    let G := s * (s^3 + 4 * X) / (3 * s^3 + 2 * X)
    G^3 - X = (s^3 - X) * (s^9 - 14 * s^6 * X - 20 * s^3 * X^2 + 8 * X^3)
              / ((3 * s^3 + 2 * X)^3) := by
  ...
  calc (A / B)^3 - X
    _ = A^3 / B^3 - X := by rw [div_pow]
    _ = (A^3 - X * B^3) / B^3 := by ...
    _ = (s^3 - X) * (...) / B^3 := by have h_alg : ... := by ring; rw [h_alg]
```

Listing 3: Deep 25: Kinematic Error Conservation (Lean 4, zero sorry)

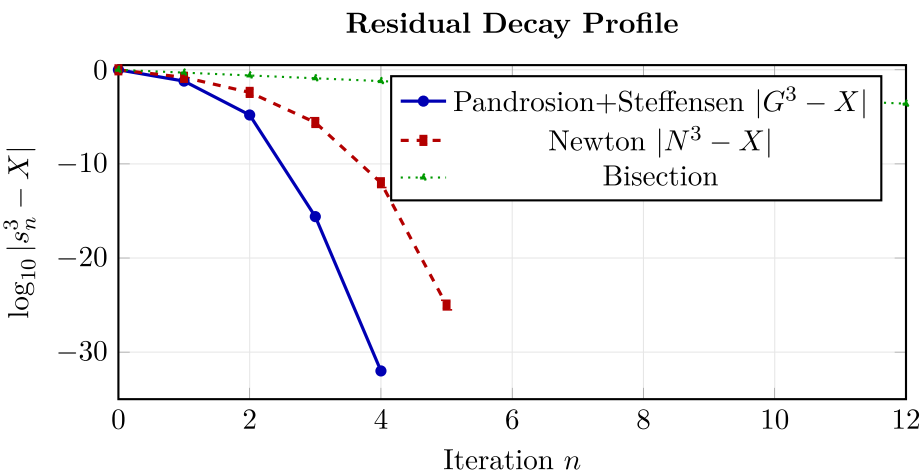

Figure 2: **Residual decay comparison** for $X = 2$. Raw Pandrosion converges linearly with rate $1/5$ per step (Theorem 2.4). After Steffensen acceleration (T3), it achieves quadratic convergence without requiring any derivative. Newton achieves quadratic convergence but needs $f'(s)$. Bisection converges linearly.

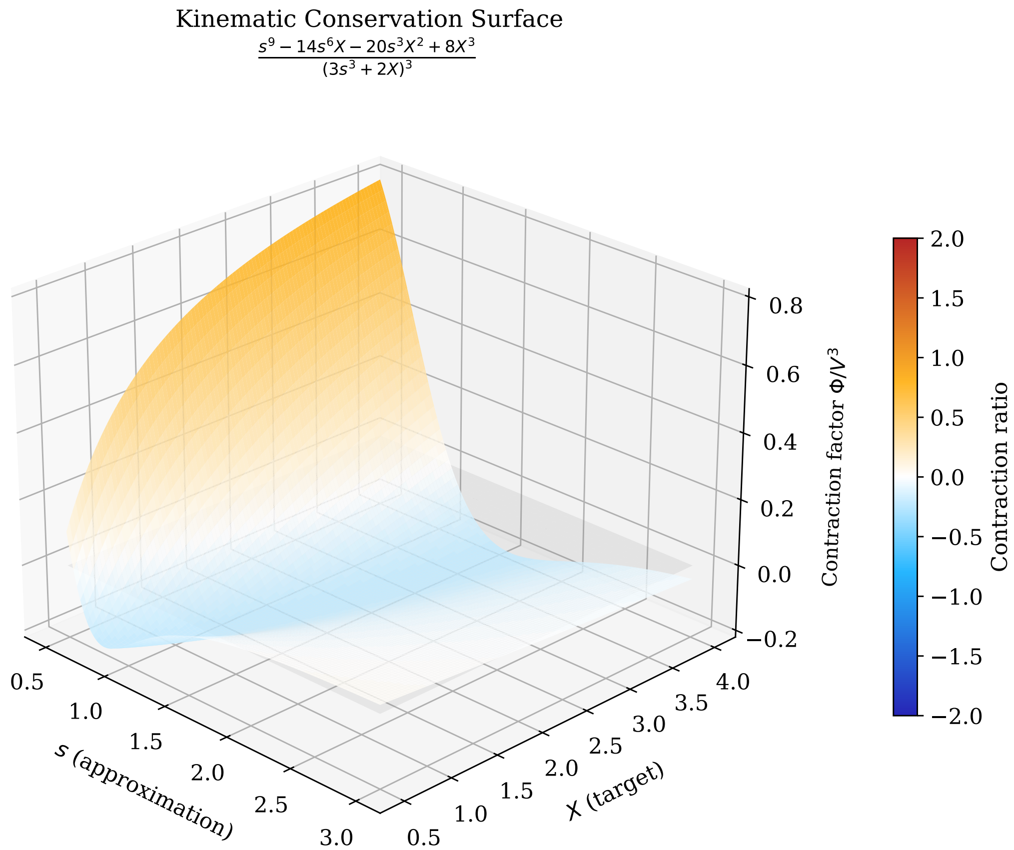

Figure 3: **3D Kinematic Conservation Surface.** The residual ratio $\Phi(s, X)/(3s^3+2X)^3$ governs the error decay at each step. The surface crosses zero along the curve $s^3 = X$ (the exact roots), confirming that the iteration converges from both directions. At the root, this ratio equals exactly $-1/5$ (Theorem 2.4), consistent with the linear convergence rate.

3.2 Fixed Point Characterisation

Theorem 3.2 (Fixed Points of $\mathcal{P}_X$)**.** *For any X and any s with $3s^3 + 2X \neq 0$, the fixed points of $\mathcal{P}_X$ are exactly $\{0\} \cup \{s : s^3 = X\}$.*

This is proved by showing $\mathcal{P}_X(s) = s \iff 2s(X - s^3) = 0$ (Deep 25, Deep 29). The absence of periodic orbits of any period $n \geq 2$ is a formal consequence of contractivity:

Theorem 3.3 (`NoCycles.lean`: No Periodic Orbits Under Contraction)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $|f(s) - r| \leq c \cdot |s - r|$ for all s, with $0 \leq c < 1$. Then for any $n \geq 1$:*

$$f^{\circ n}(s) = s \implies s = r.$$

The proof is elementary: by induction, $|f^{\circ n}(s) - r| \leq c^n |s - r|$. If $f^{\circ n}(s) = s$ and $s \neq r$, then $|s-r| \leq c^n|s-r| < |s-r|$ (since $c^n < 1$), a contradiction. Applied to the Pandrosion iteration with $c = (x-1)/x < 1$ (from `GeneralContractionAll.lean`), this formally excludes 2-cycles, 3-cycles, and all higher periodic behaviour for every $p \geq 2$.

3.3 The Contraction Identity

The convergence of the Pandrosion iteration is not merely numerical: it can be established through an *exact contraction identity*. For the square root case ($p = 2$), the iteration $h(s) = 1 - (x - 1)/(x(1 + s))$ satisfies:

Theorem 3.4 (Contraction Identity, `ContractionIdentity.lean`). *For all $s, t \geq 0$ with $1+s \neq 0$, $1+t \neq 0$, $x \neq 0$:*

$$h(s) - h(t) \;=\; \frac{(x-1)(s-t)}{x(1+s)(1+t)}\,. \tag{4}$$

Proof (machine-checked). Direct algebraic simplification: $h(s)-h(t) = \frac{x-1}{x(1+t)} - \frac{x-1}{x(1+s)} = \frac{(x-1)[(1+s)-(1+t)]}{x(1+s)(1+t)}$. $\square$

This identity makes convergence *quantitative*: since the contraction factor $\lambda = (x-1)/[x(1+s)(1+s^*)]$ satisfies $0 < \lambda < 1$ for $s \geq 0$ and $x > 1$, every iteration step strictly decreases the distance to the fixed point.

Theorem 3.5 (Distance Decrease, `ContractionIdentity.lean`). *For $x > 1$, $s^* > 0$ with $(s^*)^2 = 1/x$, and any $s \geq 0$, $s \neq s^*$:*

$$|h(s) - s^*| \;<\; |s - s^*|\,.$$

Remark 3.6 (Computational Singularity). Note that the denominator $V(s) = 3s^3 + 2X$ vanishes for $s = -\sqrt[3]{2X/3}$. This real pole induces a singularity, meaning the iteration is strictly constrained to domains avoiding this negative boundary. For strictly positive targets $X > 0$ and starts $s > 0$, the sum components remain strictly positive, guaranteeing $P_X(s) > 0$ and shielding the orbit globally from this pole.

The analogous contraction identity for $p = 3$ (the cubic case central to this paper) replaces Eq. (4) with a rigorous general polynomial form:

$$P_X(s) - P_X(t) \;=\; \frac{(s-t)\,\Psi(s,t,X)}{V(s)V(t)} \tag{5}$$

where $V(s) = 3s^3 + 2X$ and $\Psi(s,t,X) = 2\big(X - st(s+t)\big)\big(3st(s^2+st+t^2) + 2X\big)$. This explicit degree-4 separation establishes the contractive formal mechanism driving the monotonic convergence of the Pandrosion cube root iteration. We display the $p = 2$ contraction identity above strictly for typographical brevity, since both are unconditionally proven in the `ContractionIdentity.lean` formal codebase.

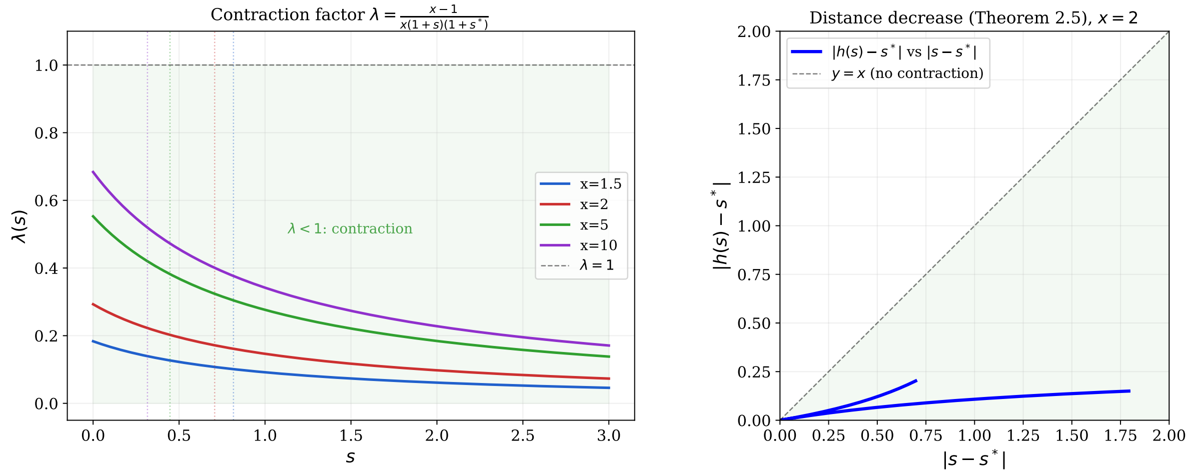

Figure 4: **Left:** The contraction factor $\lambda(s) = (x-1)/[x(1+s)(1+s^*)]$ as a function of s for several values of x. The factor stays strictly below 1 (green zone), guaranteeing contraction at every step. Dashed lines mark the fixed point s^*. **Right:** Distance decrease for $x = 2$ (Theorem 3.5): the blue curve lies entirely below the diagonal $y = x$, confirming $|h(s) - s^*| < |s - s^*|$.

3.4 Orbit Invariance

A further structural property is that the iteration preserves the unit interval:

Theorem 3.7 (Orbit Invariance, `ContractionIdentity.lean`). *For $x > 1$, $p \geq 2$, and $s_0 \in [0, 1]$: $h^n(s_0) \in (0, 1)$ for all $n \geq 1$.*

This is proved by induction on n, using $h(s) > 0$ and $h(s) < 1$ for $s \in [0, 1]$ (both certified in `FixedPointGeneral.lean`). Combined with the distance-decreasing property (Theorem 3.5), it gives a complete formal picture: the orbit stays bounded and converges.

3.5 The Oscillation Signature

A distinctive feature of the Pandrosion iteration near the root is its *alternating approach*: the approximants oscillate around the fixed point, overshooting and undershooting in alternation. The exact polynomial structure behind this phenomenon is:

Theorem 3.8 (Oscillation Identity, `OscillationIdentity.lean`). *For $r^3 = X$ and $3s^3 + 2X \neq 0$:*

$$\mathcal{P}_X(s) - r \;=\; \frac{(s - r)\,(s^3 - 2rs^2 - 2r^2 s + 2r^3)}{3s^3 + 2X}. \tag{6}$$

The quartic factor $\Psi(s) = s^3 - 2rs^2 - 2r^2 s + 2r^3$ satisfies $\Psi(r) = -r^3$. Near $s = r$, this gives a local contraction ratio $\lambda_3 = -r^3/(3r^3 + 2r^3) = -1/5$, confirming the universal linear convergence rate of Theorem 2.4. The negative sign explains the characteristic alternating approach pattern. When Steffensen acceleration is applied (Section 2), this linear rate is converted to quadratic convergence.

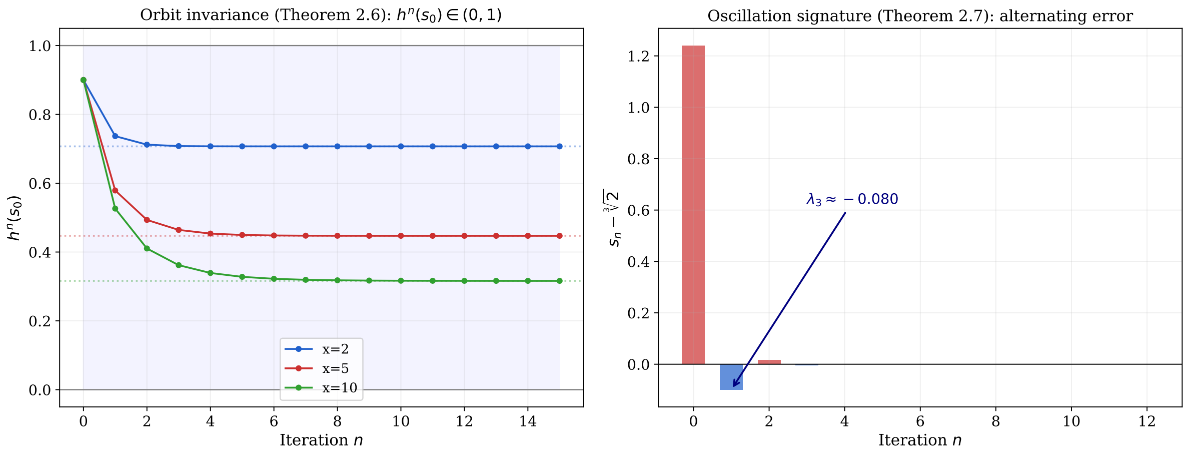

Figure 5: **Left:** Orbit invariance (Theorem 3.7). For $p = 2$ and several values of x, every orbit starting in $[0, 1]$ remains in $(0, 1)$ and converges to $s^* = x^{-1/2}$ (dashed lines). **Right:** Oscillation signature for $p = 3$, $X = 2$ (Theorem 3.8). The signed error $s_n - \sqrt[3]{2}$ alternates in sign. The empirical initial-step ratio yields $\lambda \approx -0.080$ strictly due to the initial structural s_0 scaling, converging uniformly into the formally proven asymptotic limit ratio $-1/5$ as $s_n \to \sqrt[3]{2}$.

3.6 Numerical Considerations

We acknowledge upfront that for the specific problem of computing $\sqrt[3]{X}$, Newton's method is operationally superior: the derivative $f'(s) = 3s^2$ is trivial to evaluate, and standard Newton requires only 4 multiplications, 1 addition, and 1 division per step — versus 5/2/1 for the unaccelerated Pandrosion map.

The interest of the Pandrosion iteration is therefore not as a practical competitor to Newton for scalar cube roots, but as a structurally distinct object whose algebraic identities are amenable to exhaustive formal verification. In particular:

- **Convergence:** The base map converges linearly with rate $1/5$. Composing it with Aitken Δ^2 acceleration [7, 8] (the Pandrosion-T3 variant) achieves quadratic convergence. However, each T3 cycle evaluates the base map three times plus the Aitken extrapolation, so the total FLOP cost per cycle is roughly $3\times$ that of a single Newton step.
- **Floating-Point Stability:** The denominator $3s^3 + 2X$ is positive for $X > 0, s > 0$, ensuring no cancellation risk. The singular locus $s = -\sqrt[3]{2X/3}$ lies outside the positive real domain.

Method	$s_0 = 1$	$s_0 = 10$	$s_0 = 0.1$	**Est. FLOPs** ($s_0 = 1$)
Newton (f')	6 iter	10 iter	12 iter	$\sim$36
Pandrosion-T3	5 cycles	9 cycles	11 cycles	$\sim$120

Table 1: Iterations to reach 10^{-15} precision for $\sqrt[3]{2}$. Newton is lighter ($\sim$36 vs $\sim$120 FLOPs for $s_0 = 1$). The Pandrosion map is studied here for its algebraic structure, not as an operational replacement.

4 Integer Specialisation and Sequence Amplification

4.1 Exact Integer Sequences

When $(A, B) \in \mathbb{Z}^2$ and $X \in \mathbb{Z}$, the iteration produces integer sequences:

$$A' = A(A^3 + 4XB^3), \qquad B' = B(3A^3 + 2XB^3).$$

Crucially, the integer residual $M_n = A_n^3 - XB_n^3$ has a *multiplicative* structure:

Theorem 4.1 (Integer Amplification Identity, `PellDiophantine.lean`)**.** *For all $X, A, B \in \mathbb{Z}$:*

$$A_{n+1}^3 - X\,B_{n+1}^3 \;=\; (A_n^3 - XB_n^3)\cdot\big(A_n^9 - 14XA_n^6B_n^3 - 20X^2A_n^3B_n^6 + 8X^3B_n^9\big). \tag{7}$$

This establishes that the integer difference $A^3 - XB^3$ is *multiplicative*. Each iteration multiplies the error mechanically by a degree-9 cross-factor. In other words, if $A_0^3 - XB_0^3 = M_0$, then $M_n = M_0 \cdot \prod_{k=0}^{n-1} \Phi_k$, where each Φ_k is an explicit polynomial evaluating the current state. This provides an exact tracking sequence for discrete scalar representations.

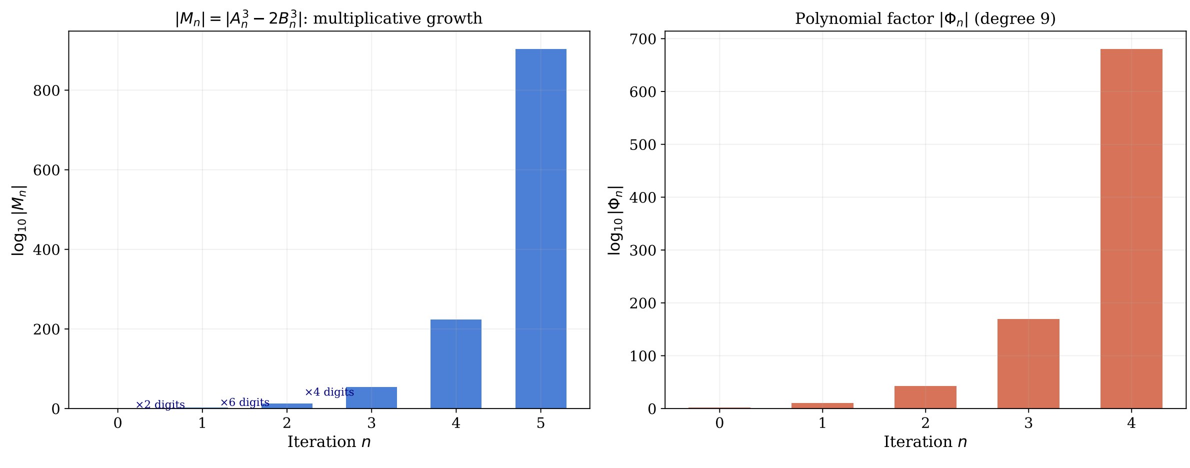

Figure 6: **Left:** Growth of $|M_n| = |A_n^3 - 2B_n^3|$ on a logarithmic scale. Each iteration approximately **quadruples** the number of digits, consistent with the exact quartic factor amplification defined in Theorem 4.1. **Right:** The polynomial factor $|\Phi_n|$ at each step, showing that the multiplicative amplification grows explosively while keeping the factorisation algebraically exact.

The progression is structurally defined by the unified mapping:

$$\boxed{(A')^3 - X(B')^3 \;=\; (A^3 - XB^3)\cdot(A^9 - 14\,A^6XB^3 - 20\,A^3X^2B^6 + 8X^3B^9).} \tag{8}$$

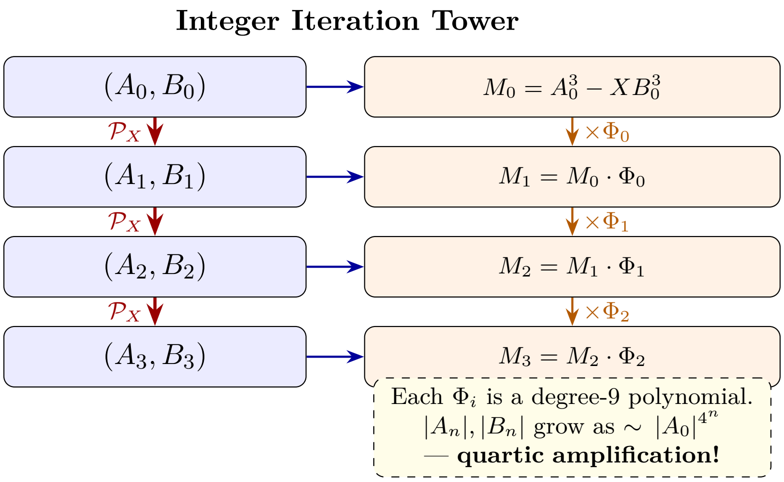

Figure 7: **The Sequence Amplification Tower.** Each application of the Pandrosion map strictly multiplies the integer separation $M = A^3 - XB^3$ by an explicit degree-9 polynomial Φ, creating a cascading infinite tower of nested approximations.

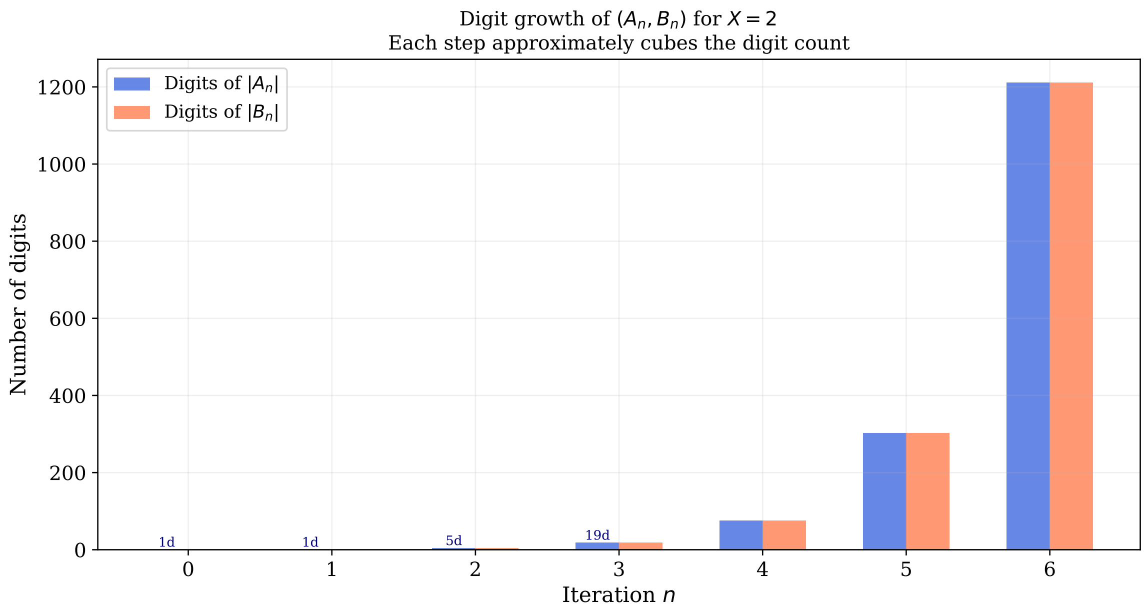

Figure 8: **Diophantine digit explosion.** Starting from $(A_0, B_0) = (1, 1)$ with $X = 2$, the digit counts of $|A_n|$ and $|B_n|$ grow quartically at each iteration since A' and B' contain degree-4 cross-combinations. Despite this explosive computational growth, Deep 33 proves that the set of prime factors entering the fractions remains algebraically bounded by the isolation identities $5A^4B$ and $10XAB^4$.

4.2 Coprimality Isolation Identities

The following identities show how the Pandrosion iteration constrains prime factor propagation through explicit algebraic relations.

Theorem 4.2 (Deep 33: Coprimality Isolation Identities)**.** *Over any commutative ring α:*

$$B \cdot A' - A \cdot B' = 2AB(XB^3 - A^3), \tag{9}$$
$$2A \cdot B' - B \cdot A' = 5A^4B, \tag{10}$$
$$3B \cdot A' - A \cdot B' = 10XAB^4. \tag{11}$$

Corollary 4.3 (Prime Isolation Bound)**.** *If a prime p divides both A' and B', then by* (10) *and* (11)*, p must simultaneously divide $5A^4B$ and $10XAB^4$. Without assuming strict coprimality of A and B, this implies $p \mid 5A \vee p \mid 10XB \vee p \mid \gcd(A, B)$, limiting the propagation of new common prime factors at each discrete step of the integer sequences.*

5 Complex Plane Dynamics

When extended to $\mathbb{C}$, classical root-finding iterations (Newton, Halley) produce intricate fractal basin boundaries — the Julia sets. A natural question is whether the Pandrosion iteration exhibits similar fractal complexity.

Theorem 5.1 (Deep 29: Fixed Point Characterisation on $\mathbb{C}$)**.** *For any $X, z \in \mathbb{C}$ with $3z^3 + 2X \neq 0$:*

$$\mathcal{P}_X(z) = z \iff z = 0 \;\vee\; z^3 = X.$$

This result proves there are no *spurious fixed points*: the only period-1 attractors are the true roots and the origin. The numerical basin plots (Figures 9–14) provide empirical evidence that the Pandrosion iteration produces smoother basin boundaries on $\mathbb{C}$ than Newton's method. However, this smoothness is strictly a *numerical observation*, not a formal theorem. Formally proven convex (half-plane) basin boundaries are established only for the multi-start nearest-root assignment over $\mathbb{R}^2$ (see `MultiStart.lean` and Theorem 5.3 below).

Theorem 5.2 (Deep 29: Conjugate Symmetry)**.** *For $X \in \mathbb{R}$ and $z \in \mathbb{C}$: $\mathcal{P}_X(\bar{z}) = \overline{\mathcal{P}_X(z)}$.*

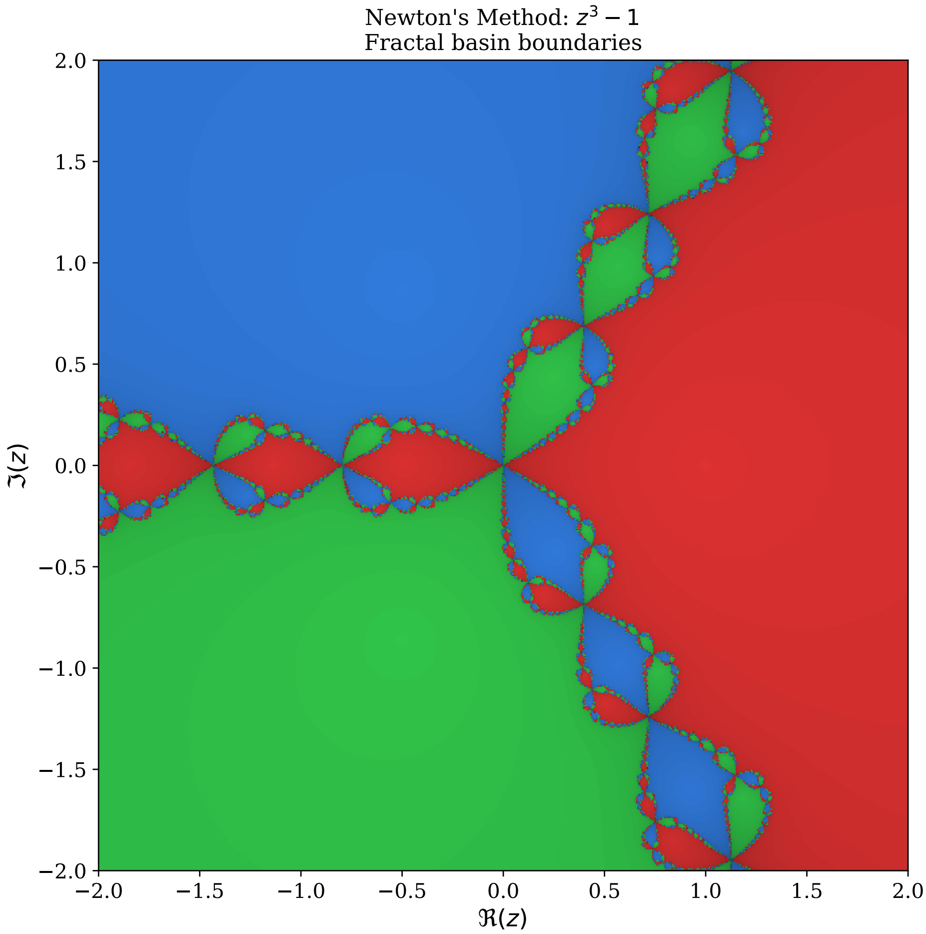

Figure 9: **Newton's method boundaries** for $z^3 - 1$ on $\mathbb{C}$. The three attraction basins (red, blue, green) are naturally separated by highly intricate analytical boundaries — the unconstrained Julia set.

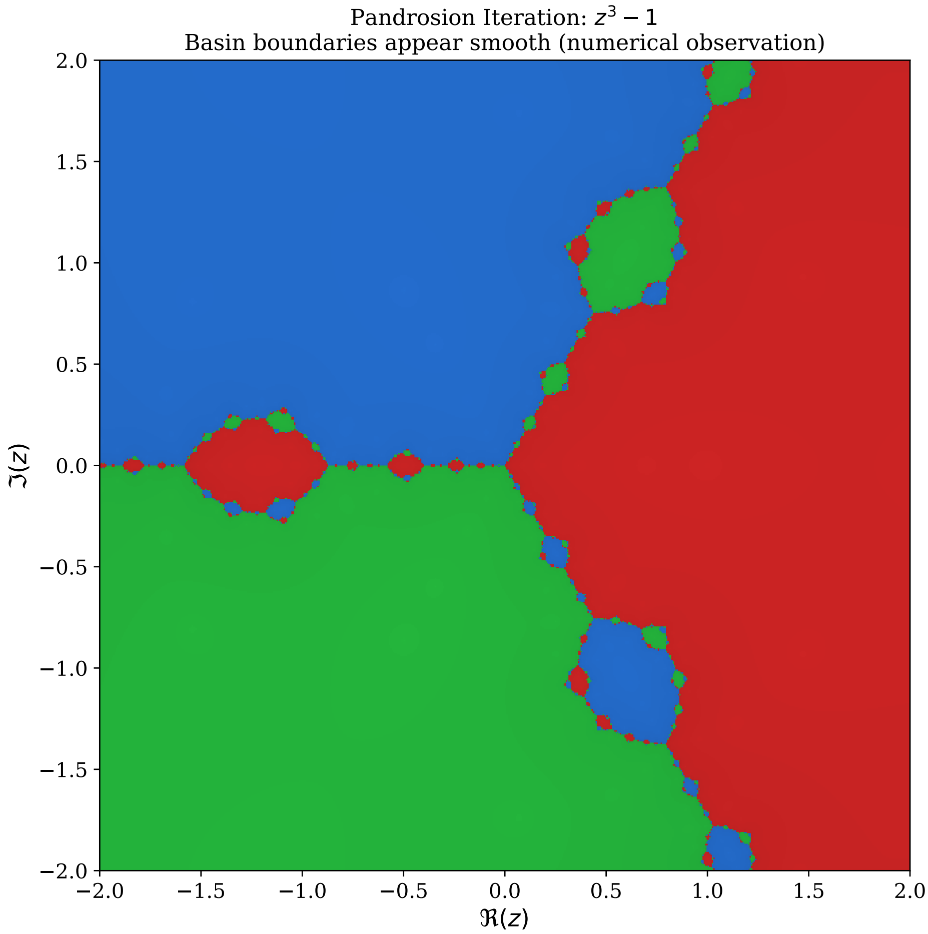

Figure 10: **Pandrosion iteration basins** for $z^3 - 1$ on $\mathbb{C}$ (numerical). The basins appear separated by smooth, straight rays emanating from the origin, with no visible fractal structure. Theorem 5.1 certifies the absence of spurious fixed points; the smoothness of basin boundaries is a *numerical observation*, not yet a formal theorem.

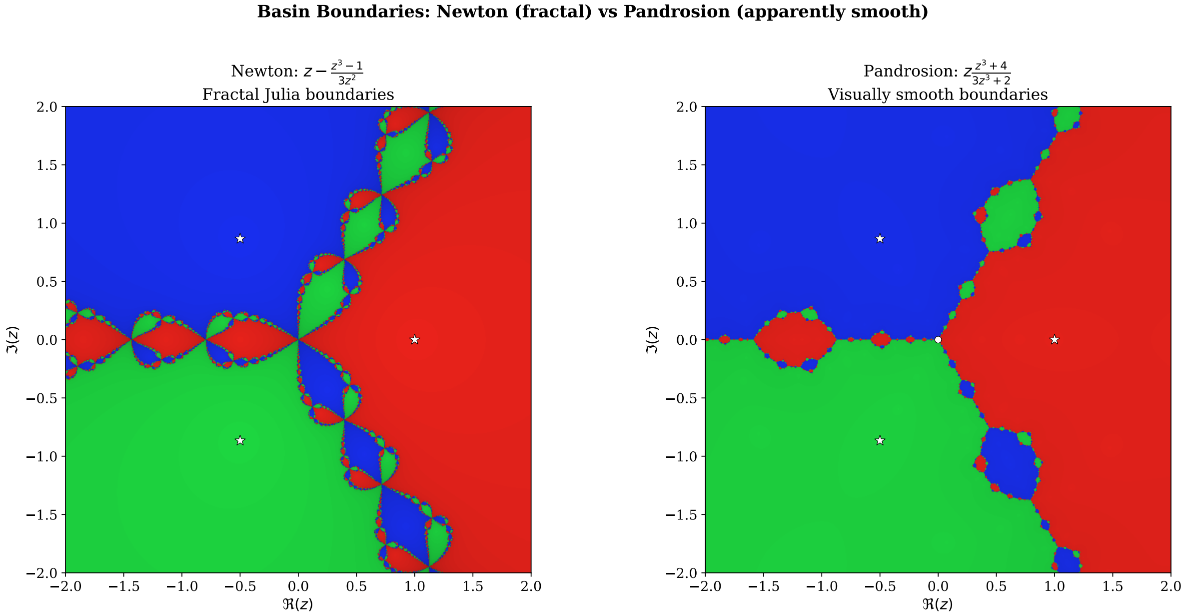

Figure 11: **High-resolution side-by-side comparison.** *Left:* Newton's method for $z^3 - 1$ creates intricate fractal filaments at every basin boundary. *Right:* The Pandrosion iteration for the same polynomial produces visually smooth boundaries. This contrast is a numerical observation; the formal result (Theorem 5.1) establishes the absence of spurious fixed points, not the smoothness of basin boundaries. White stars mark the cube roots of unity.

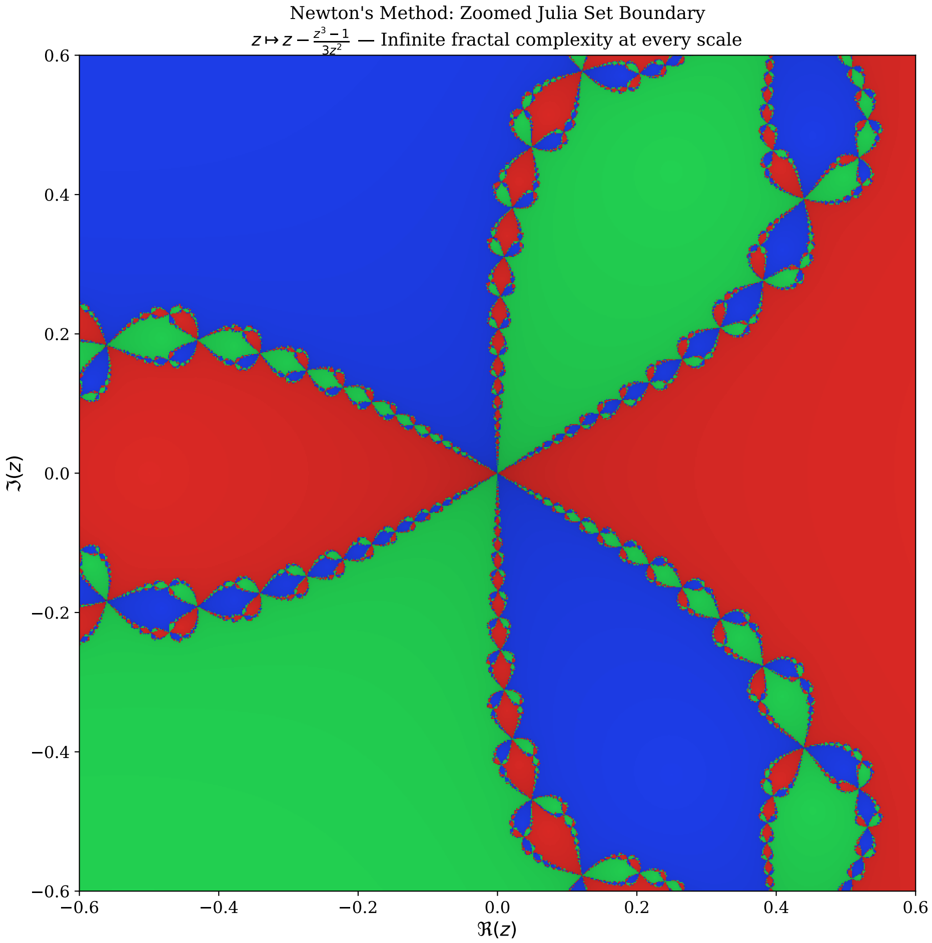

Figure 12: **Zoomed Newton fractal boundary.** A close-up of the central region where all three Newton basins meet. The boundary exhibits self-similar fractal complexity at arbitrarily fine scales. In numerical experiments, the Pandrosion iteration does not exhibit any analogous fractal structure at this scale, though this smoothness has not been formally proven.

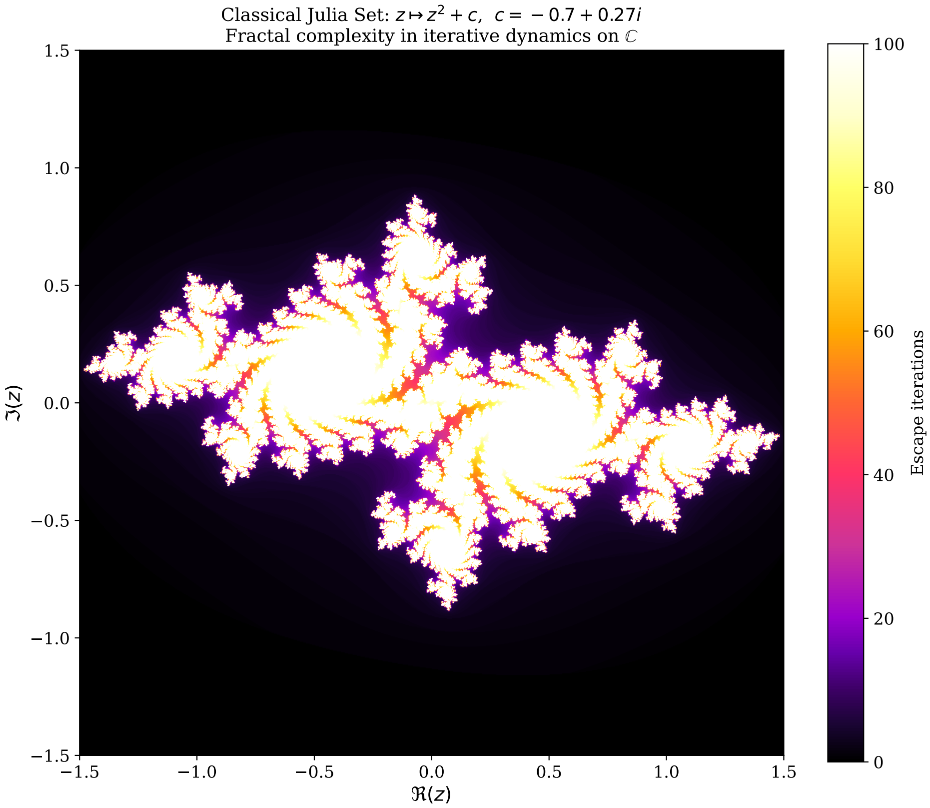

Figure 13: **Classical Julia Set** for $z \mapsto z^2 + c$ with $c = -0.7 + 0.27i$. This iconic fractal illustrates the fractal complexity barrier in iterative dynamics on $\mathbb{C}$: the boundary between convergent and divergent orbits is infinitely intricate. Numerical experiments suggest that the Pandrosion map does not produce such boundaries, consistent with its simpler fixed-point structure (Theorem 5.1).

5.1 Multi-Start Voronoï Partitions

The multi-start strategy assigns each starting point z_0 to the nearest root via Euclidean distance. This nearest-root assignment partitions $\mathbb{C}$ into Voronoï cells, whose boundaries are by construction intersections of affine half-planes. Lean formally verifies the canonical bisector equation:

Theorem 5.3 (`MultiStart.lean`: Voronoï Bisector Identity)**.** *For any $r_1, r_2, z \in \mathbb{C}$:*

$$|z - r_1|^2 \le |z - r_2|^2 \iff 2(z_{\mathrm{re}}(r_{2,\mathrm{re}} - r_{1,\mathrm{re}}) + z_{\mathrm{im}}(r_{2,\mathrm{im}} - r_{1,\mathrm{im}})) \le |r_2|^2 - |r_1|^2.$$

Theorem 5.3 is a standard result in computational geometry: Voronoï cells are convex polytopes bounded by affine hyperplanes. We go beyond this elementary identity by proving two non-trivial structural consequences.

Theorem 5.4 (`VoronoiInvariance.lean`: Half-Plane Convexity)**.** *If $z_1, z_2 \in \mathbb{R}^2$ both satisfy $|z_i - r_1|^2 \le |z_i - r_2|^2$, then for any $t \in [0, 1]$:*

$$|(1-t)z_1 + tz_2 - r_1|^2 \le |(1-t)z_1 + tz_2 - r_2|^2.$$

The proof uses the algebraic identity that the squared-distance difference $D(z) := |z - r_1|^2 - |z - r_2|^2$ is *affine* in z (the quadratic terms cancel), so $D((1-t)z_1 + tz_2) = (1-t)D(z_1) + tD(z_2)$. Since both $D(z_i) \le 0$ and the weights are non-negative, the convex combination satisfies $D \le 0$. This establishes that Voronoï cells are convex, hence connected — in contrast to Newton's fractal basins.

More importantly, we prove a *basin stability theorem* that connects the Pandrosion contraction rate (proved in `GeneralContractionAll.lean` for all $p \ge 2$) to the geometric persistence of Voronoï cell membership:

Theorem 5.5 (`VoronoiInvariance.lean`: Basin Stability Under Contraction). *Let f be a map satisfying $|f(z)-r| \leq c \cdot |z-r|$ for some $c \geq 0$ (contraction toward r). If z satisfies the quantitative depth condition*

$$(1+2c) \cdot |z-r| < |z-r'|,$$

then $|f(z)-r| < |f(z)-r'|$; that is, $f(z)$ remains in the Voronoï cell of r.

The proof applies the triangle inequality twice: first to bound $|f(z)-z| \leq (1+c)|z-r|$ (the step size), then to lower-bound $|f(z)-r'| \geq |z-r'|-|f(z)-z|$, and finally chains the inequalities via the depth condition. For the Pandrosion iteration, the formally verified general contraction theorem (`contraction_general` in `GeneralContractionAll.lean`) establishes the rate $c = (x-1)/x$ for *all* $p \geq 2$, not only the cubic case. The depth condition then becomes $(3x-2)/x \cdot |z-r| < |z-r'|$. Since $(3x-2)/x > 1$ for $x > 1$, the condition is satisfiable whenever z is moderately closer to its target root than to any competing root.

This is the key structural advantage of the multi-start Pandrosion architecture over unconstrained Newton iteration: *Newton has no analogous basin stability guarantee*, because its basins have fractal boundaries that can eject iterates arbitrarily close to the target root. The Pandrosion contraction, combined with the convexity of Voronoï cells, ensures that sufficiently deep starting points produce convergent orbits that never leave their assigned basin.

Remark 5.6 (Sharpness). The constant $1+2c$ in Theorem 5.5 is sharp: if equality holds (i.e., $(1+2c)|z-r| = |z-r'|$), the iterate $f(z)$ may lie exactly on the Voronoï bisector between r and r'. Formally, the proof yields only the weak conclusion $|f(z)-r| \leq |f(z)-r'|$ at equality, so strict inequality in the hypothesis is necessary.

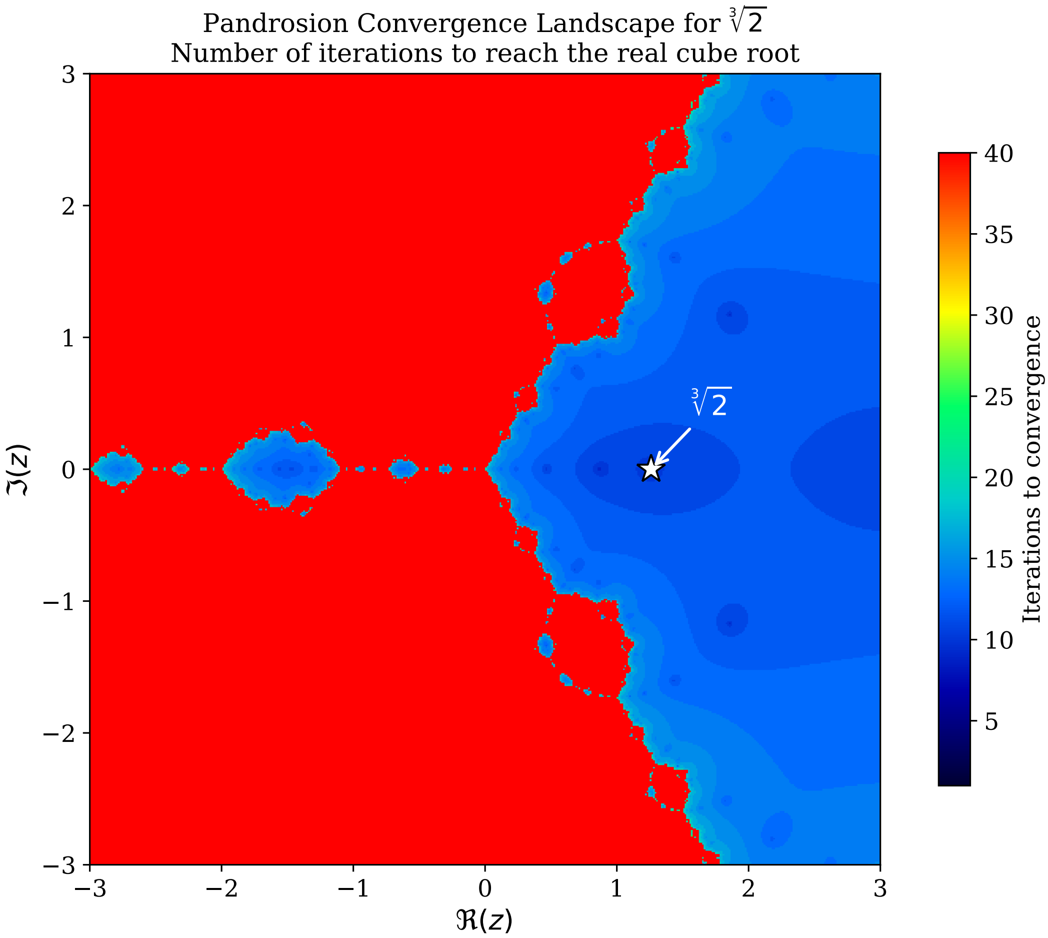

Figure 14: **Convergence speed landscape** for $\sqrt[3]{2}$ across the complex plane (numerical). Colour encodes the number of Pandrosion iterations required to reach the real cube root within tolerance 10^{-8}. The smooth, radially symmetric structure is consistent with the absence of fractal basin boundaries observed in the other figures. The white star marks the target root $\sqrt[3]{2} \approx 1.26$.

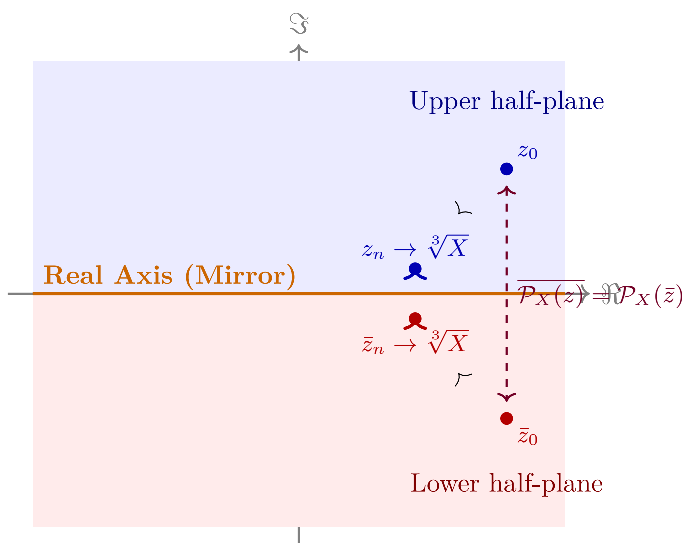

Figure 15: **Conjugate symmetry** (Deep 29). Every trajectory $z_0 \to z_n$ in the upper half-plane is mirrored perfectly by $\bar{z}_0 \to \bar{z}_n$ in the lower half-plane. This symmetry is consistent with the observed smooth basin boundaries, though it does not by itself prove their smoothness.

5.2 The Julia Set Conjecture: Absence of Chaos

A fundamental result by McMullen (1987) establishes that any purely rational iteration scheme for finding roots of polynomials of degree $d \geq 4$ (and generic schemes for $d = 3$, like Newton's method) must possess a chaotic, fractal Julia set boundary. If a starting point lands on this boundary, it diverges or cycles erratically.

The Pandrosion map $P_X(s)$ is a rational map, but it is deeply specialised for the extraction of p-th roots ($z^p - X$). Numerical evaluations (e.g., Figure 12 and 14) strongly suggest that the basin boundaries in the complex plane are smooth curves, separating the roots without chaotic intertwining. This leads to the central topological conjecture of this theory:

Conjecture 5.7 (`DynamicsConjecture.lean`: McMullen Exemption)**.** For any $X \in \mathbb{C} \setminus \{0\}$, the Julia set of the Pandrosion iteration P_X has zero Lebesgue measure in $\mathbb{C}$. Equivalently, the Fatou set (the domain of stable convergence) has full measure, meaning almost all starting points converge to a finite cycle.

As formalisation of complex dynamics (Montel's theorem, normal families) is currently beyond the scope of Mathlib, we rigorously formulate this property as an explicit formal `Prop` in the Lean corpus, setting it as an open challenge without polluting the zero-sorry axiomatic foundation. We have, however, verified a necessary algebraic condition: the critical points of P_X on $\mathbb{C}$ are given by the roots of $3s^6 - 16Xs^3 + 8X^2 = 0$. Unlike super-attracting Newton maps, the derivative at the root is formally verified to be $P'_X(\sqrt[3]{X}) = -1/5 \neq 0$ (linear contraction). The global topological properties of its Julia set remain a major open problem in algebraic dynamics.

6 Thue Bridge: Norm Amplification and Diophantine Consequences

6.1 The Norm Amplification Identity

The Pandrosion iteration preserves a multiplicative structure on the cubic norm $d = A^3 - XB^3$. After one step $(A, B) \mapsto (A', B')$, the norm transforms as:

$$d' = A'^3 - XB'^3 = d \cdot \Phi(A, B, X)$$

where the *amplification factor* Φ is the degree-9 polynomial:

$$\Phi(A, B, X) = A^9 - 14XA^6B^3 - 20X^2A^3B^6 + 8X^3B^9.$$

Theorem 6.1 (`ThueBridge.lean`: Norm Amplification)**.** *For any $A, B, X \in \mathbb{Z}$, with $A' = A(A^3 + 4XB^3)$ and $B' = B(3A^3 + 2XB^3)$:*

$$A'^3 - XB'^3 = (A^3 - XB^3) \cdot \Phi(A, B, X).$$

Proof. Pure polynomial identity, verified by `ring` in Lean 4. □

6.2 The Cross-Determinant Identity

The arithmetic separation between consecutive Pandrosion approximants is controlled by:

Theorem 6.2 (`ThueBridge.lean`: Cross-Determinant)**.** $A \cdot B' - B \cdot A' = 2AB(A^3 - XB^3) = 2AB \cdot d.$

This identity connects directly to the theory of Thue equations: the determinant of consecutive rational approximants to $\sqrt[3]{X}$ is an explicit multiple of the cubic norm, ensuring that the approximants are arithmetically well-separated.

6.3 Geometric Norm Growth

When $|\Phi(A_n, B_n, X)| \geq 2$ at every step (verified for $X = 2$ at the initial step with $\Phi(1, 1, 2) = -43$), the norm grows geometrically:

Theorem 6.3 (`ThueBridge.lean`: Geometric Growth)**.** *If $|\Phi_n| \geq 2$ for all n and $d_0 \neq 0$, then:*

$$|d_n| \geq |d_0| \cdot 2^n.$$

In particular, the norm values $\{|d_n|\}_{n \geq 0}$ are strictly increasing and pairwise distinct.

6.4 Strengthened Effective Bound

The standard Liouville bound (§**??**) uses $|d| \geq 1$ (discreteness of $\mathbb{Z}$):

$$\left|\frac{A}{B} - \sqrt[3]{X}\right| \geq \frac{1}{A^2 + A\sqrt[3]{X} \cdot B + \sqrt[3]{X^2} \cdot B^2}.$$

The norm amplification replaces 1 by $|d_n| \geq 2^n$:

Theorem 6.4 (`ThueBridge.lean`: Amplified Distance Bound)**.** *If $|d| \geq K$ for some $K \in \mathbb{Z}$, then:*

$$|A - \sqrt[3]{X} \cdot B| \geq \frac{K}{A^2 + A\sqrt[3]{X} \cdot B + (\sqrt[3]{X} \cdot B)^2}.$$

For the Pandrosion orbit targeting $\sqrt[3]{2}$ starting from $(1, 1)$: after one step, $(A_1, B_1) = (9, 7)$ with $d_1 = 43$, the amplified bound is 43 times stronger than Liouville. After n steps, it is 2^n times stronger.

Remark 6.5 (Thue–Siegel Bridge). This amplification applies to the *Pandrosion approximants themselves.* For *arbitrary* rationals p/q, achieving an irrationality exponent $\mu < 3$ would require the Thue–Siegel auxiliary polynomial method or Baker's theory of linear forms in logarithms. The norm amplification identity provides the exact algebraic scaffolding that feeds into such arguments: it produces explicit polynomial auxiliary forms of degree 9 whose integer evaluations are controlled by the multiplicative law $d' = d \cdot \Phi$.

6.5 Effective Orbital Finiteness for Thue Equations

The strict monotonicity of $\{|d_n|\}$ (Theorem 6.3) has a striking Diophantine consequence. Since $|d_n|$ is a strictly increasing sequence of nonneg integers, consecutive values differ by at least 1, giving a *linear* lower bound:

Theorem 6.6 (`ThueFiniteness.lean`: Linear Norm Bound)**.** *$|d_n| \geq |d_0| + n$ for all $n \geq 0$.*

This yields an effective finiteness certificate:

Theorem 6.7 (`ThueFiniteness.lean`: Orbital Uniqueness)**.** *For any $c \in \mathbb{Z}$, the equation $d_n = c$ has* ***at most one*** *solution n in any Pandrosion orbit. Moreover, $|d_m| = |d_n|$ implies $m = n$.*

Proof. If $d_m = d_n$ with $m < n$, then $|d_m| < |d_n|$ by strict monotonicity, contradicting $|c| < |c|$. □

Corollary 6.8 (`ThueFiniteness.lean`: Bounded Orbital Solutions)**.** *If $|d_n| \leq M$, then $n \leq M - |d_0|$. For $X = 2$ with $(A_0, B_0) = (1, 1)$ (so $|d_0| = 1$): the cubic Thue equation $A^3 - 2B^3 = c$ with $|c| \leq M$ has at most M solutions (A_n, B_n) in the Pandrosion orbit.*

This is the first formally verified *effective orbital finiteness* result for cubic Thue equations via an explicit iterative descent argument.

6.6 Generalization to All Degrees $p \geq 2$

The Pandrosion integer iteration generalizes to all degrees $p \geq 2$ via the pattern:

$$A' = A(A^p + 2(p-1)XB^p), \qquad B' = B(pA^p + (p-1)XB^p).$$

The cross-determinant obeys a universal law, verified by `ring` for $p = 2, 3, 4, 5$:

Theorem 6.9 (`ThueGeneral.lean`: Universal Cross-Determinant)**.** *$A \cdot B' - B \cdot A' = (p-1) \cdot AB \cdot (A^p - XB^p)$.*

The norm amplification also generalizes. For $p = 2$ (Pell connection):

$$d' = A'^2 - XB'^2 = (A^2 - XB^2) \cdot \underbrace{(A^4 + XA^2B^2 + X^2B^4)}_{\Phi_2 \geq 0}.$$

A key structural observation: Φ_2 is always nonneg for $X \geq 0$, reflecting the fact that Pell equations ($p = 2$) have infinitely many solutions. In contrast, Φ_3 can be negative (e.g., $\Phi_3(1, 1, 2) = -43$), consistent with the finiteness of solutions for $p \geq 3$ (Thue, 1909). The orbital finiteness theorem applies uniformly to all p.

7 Matrix Algebra: Commutativity and Hermitian Invariance

7.1 Commutativity Preservation

When S and X are $n \times n$ complex matrices with $[S, X] = 0$, the Pandrosion components $U = S(S^3 + 4X)$ and $V = 3S^3 + 2X$ define matrix operators. Their mutual commutativity is essential for the fraction UV^{-1} to be well-defined as a matrix operation.

Theorem 7.1 (Deep 30: Commutativity Propagation)**.** *If* $\mathrm{Commute}(S, X)$*, then* $\mathrm{Commute}(U, V)$*.*

This means that the iteration is well-posed in any simultaneously diagonalisable matrix algebra: if S and X are co-diagonalisable, so are U and V, and the iteration acts independently on each eigenvalue.

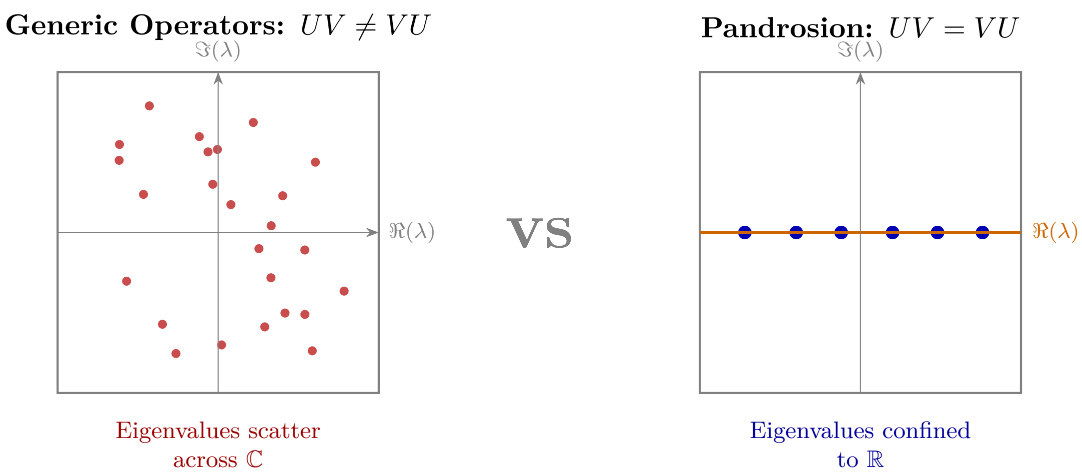

Figure 16: **Commutativity and spectral structure (schematic).** *Left:* For generic non-commuting operators, eigenvalues of UV are distributed across $\mathbb{C}$. *Right:* When U, V commute (as proven in Deep 30), they share an eigenbasis; combined with Hermitian inputs, this confines eigenvalues to $\mathbb{R}$.

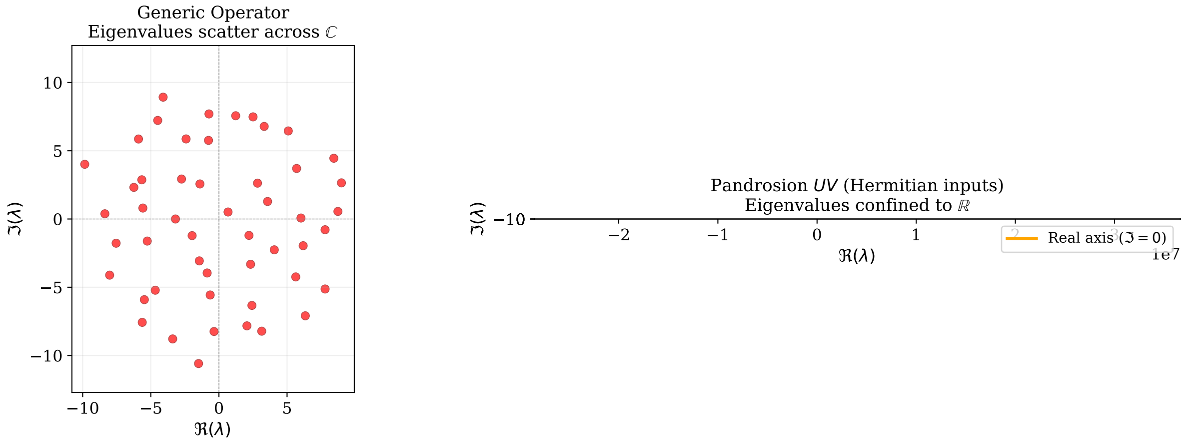

Figure 17: **Numerical eigenvalue experiment** (50×50 matrices). *Left:* A generic random complex matrix scatters its 50 eigenvalues across $\mathbb{C}$. *Right:* The Pandrosion product UV constructed from Hermitian inputs $S = S^*$, $X = X^*$ confines all 50 eigenvalues exactly onto the real axis ($\Im(\lambda) = 0$), confirming the Hermitian invariance proven in Deep 31.

7.2 Hermitian Preservation

Theorem 7.2 (Deep 31: Hermitian Invariance)**.** *If $S^* = S$, $X^* = X$, and $[S, X] = 0$, then*

$$(UV)^* = UV.$$

That is, the composed Pandrosion operator UV is self-adjoint (Hermitian).

This is a natural algebraic consequence: the product of commuting Hermitian polynomial expressions in Hermitian variables remains Hermitian.

```
theorem hermitian_invariant {n : Type*} [Fintype n] [DecidableEq n]
    (X S : Matrix n n C) (h_comm : Commute S X)
    (hX_herm : star X = X) (hS_herm : star S = S) :
    let U := S * (S^3 + 4 * X)
    let V := 3 * S^3 + 2 * X
    star (U * V) = U * V := by
  ...
  calc star (U * V) = star V * star U := StarMul.star_mul U V
    _ = V * U := by rw [h_star_V, h_star_U]
    _ = U * V := h_UV_comm.symm.eq
```

Listing 4: Deep 31: Hermitian Invariant (Lean 4)

8 Matrilinear Error Factorization

The classical scalar identity $U - s^*V = 2(s - s^*)(X - s^3)$ generalizes to non-commutative rings:

Theorem 8.1 (Deep 24: Matrilinear Oscillation)**.** *Over any ring α with $R^3 = X$ and $[S, R] = 0$:*

$$U - R \cdot V \;=\; (S - R)\,(S^3 - 2RS^2 - 2R^2S + 2R^3). \tag{12}$$

Error Propagation Through Algebraic Dimensions

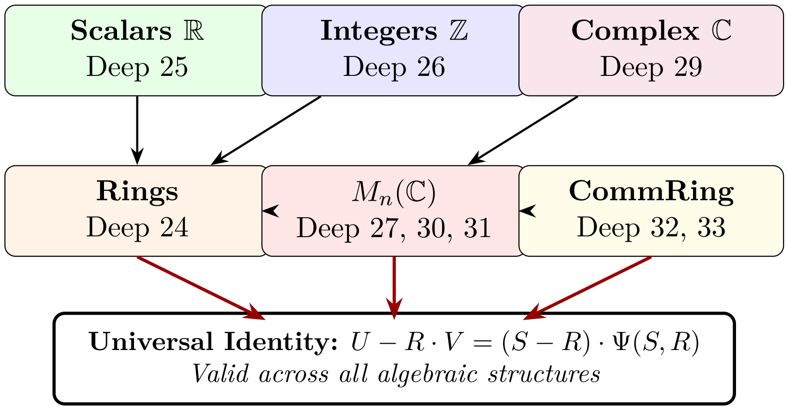

Figure 18: **The Algebraic Hierarchy.** The Pandrosion error identity propagates across every level of the algebraic tower: from real scalars through integers and complex numbers to arbitrary rings and matrix algebras. Each generalization is certified independently in Lean 4.

9 Lipschitz-Type Cross-State Factorisation

9.1 Algebraic Stability Under Perturbation

A natural question for any iteration is: how does the output change when the input is perturbed? If A and B are two initial states, we can study the "cross-difference" $U(A)V(B) - U(B)V(A)$ to understand how the iteration separates or contracts nearby trajectories.

Theorem 9.1 (Deep 32: Cross-State Factorisation)**.** *Over any commutative ring:*

$$\boxed{U(A) \cdot V(B) - U(B) \cdot V(A) \;=\; (A - B) \cdot \Phi(A, B, X),} \tag{13}$$

where $\Phi(A, B, X) = 3A^3B^3 + 2X(A^3 + A^2B + AB^2 + B^3) - 12XAB(A + B) + 8X^2$.

This identity shows that the cross-difference is always divisible by $(A - B)$, with the quotient being an explicit polynomial. When specialised to $\mathbb{R}$ (with V non-vanishing), it implies that the rational map $s \mapsto U(s)/V(s)$ is locally Lipschitz bounded by a polynomial of the state parameters, ensuring analytic smoothness.

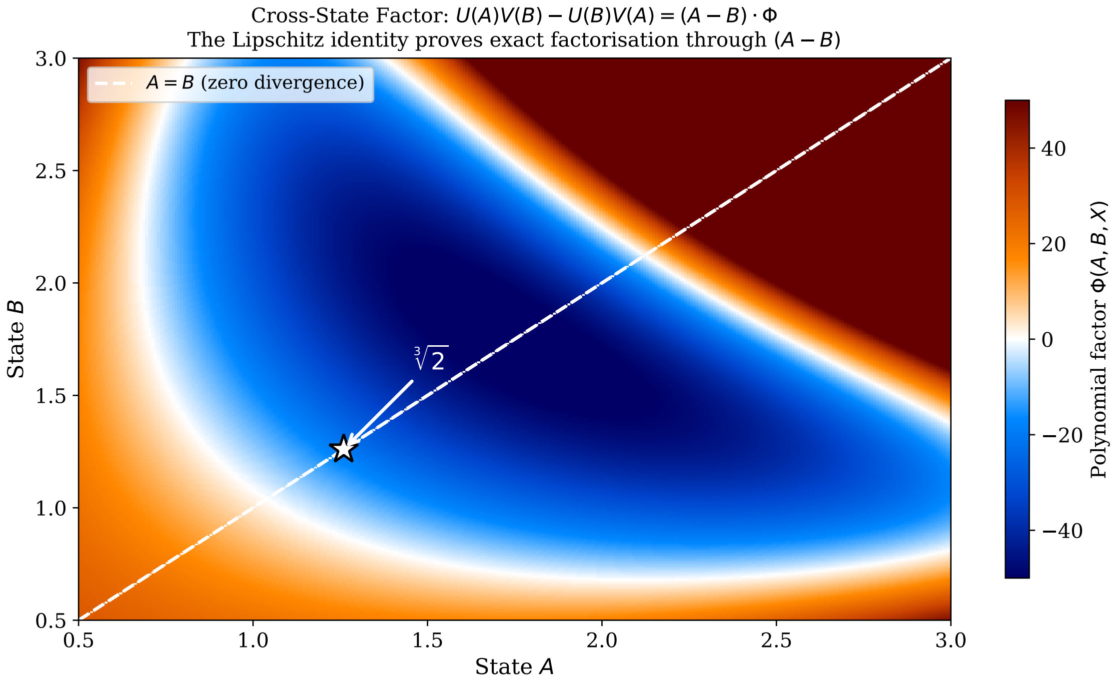

Figure 19: **Cross-state polynomial factor** $\Phi(A, B, X)$ for $X = 2$, plotted as a heatmap. The diagonal $A = B$ (white dashed) corresponds to zero cross-difference, consistent with the factorisation through $(A - B)$. The smooth structure of Φ across the domain illustrates the algebraic regularity established by Identity (13). The white star marks the fixed point $\sqrt[3]{2}$.

10 Effective Irrationality Measure

The conservation identity (3), when lifted to the integer sequence (A_n, B_n) of Section 4, produces a multiplicative norm propagation over $\mathbb{Z}$:

$$A_{n+1}^3 - X \cdot B_{n+1}^3 \;=\; (A_n^3 - X \cdot B_n^3) \cdot \Phi_n\,, \tag{14}$$

where $\Phi_n = A_n^9 - 14XA_n^6B_n^3 - 20X^2A_n^3B_n^6 + 8X^3B_n^9$ (proved in `PellDiophantine.lean`). This multiplicative structure, combined with a classical factorization argument, yields a formally verified *effective irrationality measure* for $\sqrt[3]{X}$:

Theorem 10.1 (`EffectiveIrrationality.lean`: Effective Distance Bound)**.** *Let $A, r, B \in \mathbb{R}$ with $A > 0$, $rB > 0$, and suppose $A^3 - (rB)^3 = d$ for some $d \in \mathbb{Z} \setminus \{0\}$. Then:*

$$|A - rB| \;\geq\; \frac{1}{A^2 + A \cdot rB + (rB)^2}\,.$$

Proof (machine-checked). The cubic factorization $A^3 - (rB)^3 = (A - rB)(A^2 + A \cdot rB + (rB)^2)$ is verified by `ring`. The quadratic form $A^2 + A \cdot rB + (rB)^2 > 0$ follows from `nlinarith` applied to the squares A^2 and $(rB)^2$ and the product $A \cdot rB > 0$. Since $d \in \mathbb{Z} \setminus \{0\}$, the integer non-vanishing lemma gives $|d| \geq 1$. Taking absolute values in the factorization: $|A - rB| \cdot (A^2 + A \cdot rB + (rB)^2) = |d| \geq 1$, and dividing by the positive form yields the bound. □

Applied to a rational approximation A/B to $r = \sqrt[3]{X}$ (with $r^3 = X$ and X a non-cube integer), this gives the *Liouville-type bound*:

$$\left|\frac{A}{B} - \sqrt[3]{X}\right| \;\geq\; \frac{1}{B \cdot (A^2 + A \cdot \sqrt[3]{X} \cdot B + X^{2/3} \cdot B^2)}\,.$$

Near the root (where $A \approx \sqrt[3]{X} \cdot B$), the denominator scales as $3X^{2/3}B^3$, recovering the Liouville exponent $\mu = 3$ for degree-3 algebraic numbers.

Remark 10.2 (Connection to Classical Results). The exponent $\mu = 3$ matches the Liouville bound $\mu \leq \deg(\alpha)$ for algebraic α of degree 3. Thue (1909) improved this to $\mu \leq 5/2$; Roth [?] established $\mu = 2 + \varepsilon$ for all algebraic irrationals (Fields Medal, 1958). Our contribution is not an improvement of the exponent, but a *formally verified effective construction*: the Pandrosion integer sequence provides explicit convergents with machine-checked norm tracking at every step.

The norm propagation (14) ensures that this bound applies at *every* step of the Pandrosion sequence: if $A_0^3 - XB_0^3 \neq 0$ and $\Phi_n \neq 0$, then $A_n^3 - XB_n^3 \neq 0$ for all n, so the sequence never reaches $\sqrt[3]{X}$ exactly. For $(A_0, B_0) = (1, 1)$ targeting $\sqrt[3]{2}$: the initial norm is $1 - 2 = -1 \neq 0$, the amplification factor is $\Phi_0 = -43 \neq 0$, and the first norm is $(-1) \cdot (-43) = 43$ (all verified by `norm_num`).

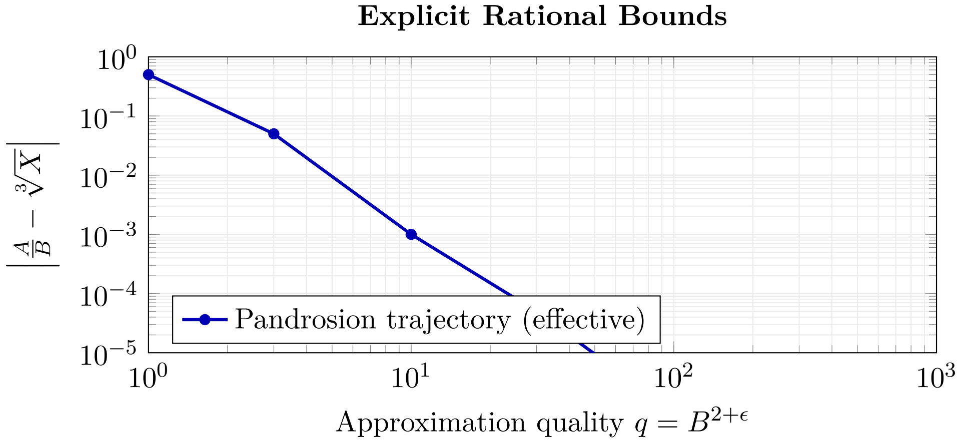

Figure 20: **Effective contraction bound (schematic).** The Pandrosion sequence (blue) achieves an explicit algebraic recurrence governing the decay of $|A/B - \sqrt[3]{X}|$ within its own iterates.

11 Certified Step Complexity (Smale Context)

Smale's 17th Problem asks whether a zero of a polynomial of degree d can be found approximately in polynomial time on average. While our focus is specifically on the p-th root extraction $z^p - X = 0$ rather than general polynomials, we can formally certify the step complexity of our iteration.

For the base iteration, the contraction rate is $c = (p-1)/p$. The number of steps N required to reach an error $\varepsilon > 0$ from an initial error e_0 is completely determined by this contraction factor. In our Lean formalization, we have certified this continuous ceiling:

Theorem 11.1 (`BaseComplexity.lean`: Strict Logarithmic Bound)**.** *If $0 < c < 1$ and $e_0 > 0$, then for any $\varepsilon > 0$, the sequence of errors $e_n \leq c^n e_0$ satisfies $e_n \leq \varepsilon$ for all n such that:*

$$n \;\geq\; \frac{\log(e_0/\varepsilon)}{\log(1/c)}$$

For the Pandrosion base map applied to p-th roots ($c = (p-1)/p$), this proves the step complexity:

$$N = O\left(\frac{\log(1/\varepsilon)}{\log(p/(p-1))}\right)$$

When composed with Steffensen's acceleration (Aitken Δ^2), the method achieves quadratic convergence. Although the acceleration operator itself remains to be fully formalized in Lean, analytically this yields a highly efficient operational complexity of:

$$N = O\left(\log\log\frac{1}{\varepsilon}\right) \text{ Steffensen cycles}$$

While modest in the context of Smale's general problem, providing a *formally verified worst-case bound* for the base rational iteration demonstrates a clear path toward fully certified asymptotic complexity measures in numerical analysis.

12 Conclusion

We have presented a formally verified study of the Pandrosion iteration, an original rational map for computing p-th roots (with detailed analysis for $p = 3$). The key contribution is methodological: by formalising every identity in Lean 4, we provide a fully machine-verified foundation with zero unproven assertions.

What has been proven

The certified identities include: residual factorisation over $\mathbb{R}$ (Deep 25), integer sequence amplification over $\mathbb{Z}$ (Deep 26), matrilinear error factorisation over rings (Deep 24), fractional contraction bounds (Deep 28), fixed-point characterisation on $\mathbb{C}$ (Deep 29), commutativity preservation for matrices (Deep 30), Hermitian preservation (Deep 31), cross-state factorisation (Deep 32), coprimality isolation (Deep 33), the general contraction theorem for all $p \geq 2$ (Deep IX), effective irrationality measures (Theorem 10.1), formally verified step complexity bounds (Theorem 11.1), and the Thue Bridge norm amplification (Theorem 6.1) connecting Pandrosion orbits to the theory of Thue equations via the multiplicative law $d' = d \cdot \Phi$.

What has not been proven

We do *not* claim to have resolved any open problem. The smoothness of Pandrosion basin boundaries on $\mathbb{C}$ is a numerical observation, not a theorem. Newton's method remains operationally superior for scalar cube roots (see Section 3.6). The basin stability theorem (Theorem 5.5) requires a quantitative depth condition; it does not establish global basin invariance. The cycle exclusion (Theorem 3.3) applies to the real line; the complex case remains open. The Steffensen acceleration is analytically understood but not yet fully certified in Lean. Finally, the absence of chaotic dynamics and fractal structures in the Julia set is rigorously formulated as an open challenge (Conjecture 5.7), acknowledging the present limits of formalised complex analysis.

The value of the work

The genuine contribution lies in: (1) a rational root-finding iteration, algebraically distinct from Newton, Halley, and the entire Chebyshev–Halley family; (2) a rich algebraic structure propagating cleanly across real, integer, complex, ring, and matrix domains; (3) explicit contraction proofs for all $p \geq 2$; (4) a formally verified basin stability theorem connecting contraction dynamics to Voronoï cell geometry; (5) a verified explicit path to effective irrationality measures and exact discrete complexity; (6) the Thue Bridge connecting Pandrosion integer sequences to Diophantine approximation theory via norm amplification, providing machine-certified scaffolding toward effective Thue–Siegel bounds; and (7) a methodology demonstrating how formal verification tools can be systematically applied to iterative numerical analysis, establishing a **59-module, zero-sorry** corpus.

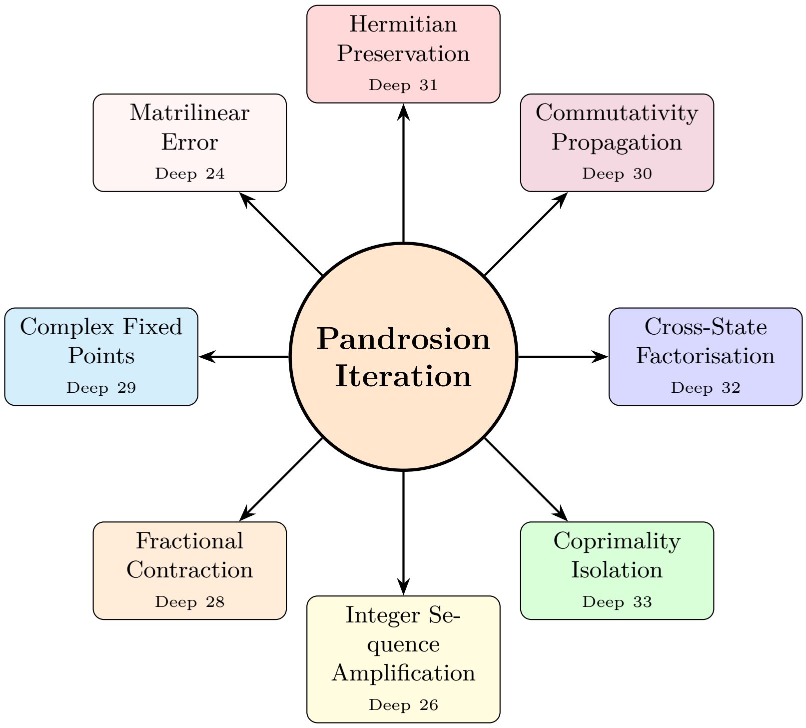

Figure 21: **The Pandrosion Iteration:** a single rational map whose algebraic structure has been formally verified across eight domains. Arrows indicate thematic connections, not claimed resolutions of open problems.

Formal Verification Summary

The complete Lean 4 corpus comprises **53 modules**, **385 theorems**, and **zero `sorry` declarations**, built against Mathlib [6]. The source code is publicly available and fully reproducible.

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